

ADAPTIVE HYBRID DEEP LEARNING FOR DAILY FINANCIAL TIME-SERIES FORECASTING USING CEEMDAN, INFORMER, AND LSTM

Abstract

This study proposes a hybrid deep learning framework for financial time-series forecasting using CEEMDAN, Informer, and LSTM models. Seven forecasting models, including standalone and CEEMDAN-based variants, are evaluated under identical experimental conditions using RMSE, MAE, MAPE, R^2 , and directional accuracy. Experimental results demonstrate that CEEMDAN-based models consistently outperform standalone models due to improved noise reduction and feature extraction. Among all models, the CEEMDAN-Informer-LSTM hybrid model achieves the best performance with RMSE 2.3529, MAE 1.8380, MAPE 0.8052, and R^2 0.9929. The framework is further extended to an adaptive rolling forecasting system incorporating volatility-based dynamic weighting, resulting in improved predictive accuracy and directional performance without requiring retraining. The results are directly derived from experimental outputs, ensuring consistency between implementation and reported findings.

Keywords: stock price prediction, CEEMDAN, LSTM, Transformer, Informer, hybrid deep learning, adaptive forecasting, rolling forecasting, financial time-series forecasting, close-price prediction.

1. Introduction

Stock prediction has become one of the most important research areas in the field of financial analytics owing to the increased complexity, ambiguity and instability of the stock markets. General character of stock prices changes, which is nonlinear and non-stationary, does not tend to be modeled by conventional statistical methods such as ARIMA and regression-based models. As a result, deep learning methods have been of great interest owing to their ability to learn latent time dynamics, and long-term association to financial time series data. Long Short-Term Memory (LSTM), Transformer, and Informer are models that can be extensively applied in forecasting tasks because they have the capability of high sequence learning. However, deep learning models in isolation tend to perform poorly with regard to predictions in situations where such markets data are disjointed and unstable. To address this weakness, researchers have started integrating signal decomposition techniques such as Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) with the deep learning systems. CEEMDAN helps to deconstruct more complex stock price signals into unsophisticated intrinsic mode forms that are more effective at extracting features and reduce noise before prediction. Some recent studies have proved the models of CEEMDAN to exhibit strong positive influence on the forecasting performance in the stock market. One such example is the use of CEEMDAN, ensemble learning and recurrent neural network, in the prediction of the price of carbon and stock indexes using a more accurate and stable prediction [1], [2]. Similarly, hybrid models using CEEMDAN which also utilize mechanisms of attention and BiLSTM have been reported to be more effective on nonlinear patterns of the market and increase the accuracy of the forecasts [3]. The outcomes imply that the potential of deep learning models can be enhanced with the assistance of preprocessing stock price data with CEEMDAN and more credible prediction performance in the dynamic financial environment may be obtained [4].

Recent advancements in stock market forecasting have been guided towards the combination of multiple deep learning models in order to address the disadvantages of a single-model based stock market prediction. LSTM networks are applicable in learning sequential dependencies, but it may not be practical in learning information over a long distance. The problem is addressed by using transformer-based models, through their attention mechanisms that enable the model to focus on the most meaningful aspects of the input sequence. It is more efficient too as informer models are less complex to compute and could

predict long runs of financial sequences faster. Although these models have proven successful at the individual level, researchers have found that hybrid models combining decomposition and more advanced deep learning architectures are more effective in producing high level of prediction accuracy as compared to single approaches. Hybrid models are armed with the capabilities of a number of different architectures and because it allows them to take advantage of fluctuation in the short run and the long-run market tendencies. Recent studies have shown the efficacy of the CEEMDAN in combination with LSTM and Transformer and other systems of ensemble learning could improve the forecast quality of stock price, price of coal and even the carbon trading market [5], [6]. In addition, more advanced hybrid systems that consider the application of LSTM, Transformer attention, sentiment analysis, and federated learning have also shown the amazing performances when it comes to stock market forecasting because they can process a wide scope of data input simultaneously [7]. Multiphase frameworks based on variational Autoencoders, Transformer models, as well as LSTM network-based ensemble frameworks are also proposed to improve the stability and robustness of stock prediction frameworks [8]. These studies suggest that a combination of signal breaking down with numerous deep learning models can create a more reliable forecasting framework that can recalibrate itself to the fresh market conditions that emerge in a very rapid manner. Therefore, hybrid CEEMDAN-based designs prove to be influential trend in existing financial forecasting studies [9].

The study focuses on forecasting the daily close-price series using the experimental framework, where the Close variable is used as the prediction target. As successful as the hybrid deep learning methods were getting, more specific and dynamic models of forecasting should be offered that would react to the changing state of affairs in the market at the time. The financial markets are highly volatile and the stock prices trend can considerably shift due to economic phenomena, sentiment of investors, geopolitical and uncertainty on the market. In most instances, prediction by means of a fixed prediction model is inaccurate and as the market adapts over time the model does not predict either. To overcome this task, the adaptive forecasting models have been suggested which can dynamically adjust the model parameter and improve the prediction. Adaptive models are useful in addressing the concept drift and dynamic tendencies of the stock market unlike the static ones. Moreover, CEEMDAN decomposition, coupled with advanced prediction models, allows the model to remove noises and useful signals, improving the overall predictive power. The new studies indicate that decomposed-based hybrid models

are superior to the conventional forecasting methods in the metrics of RMSE, MAE, MAPE and R2 [2], [3]. Even hybrid time series forecasting models (i.e., a combination of work of econometric models, machine learning models and deep learning models) have been found to do better than independent model models [9]. Similarly, techniques of decomposition, such as SVMD with LSTM have been also shown to yield positive outcomes so far as predicting the stock market in terms of eliminating noise and enhancing predictive analytics on trends acquisition [10]. As was planned in these advances, the present research study is planned to evaluate the effectiveness of the independently applied and CEEMDAN-based deep learning-based stock price predictors. The study is also compared in terms of the structures of both the static and adaptive forecasting frameworks as a by-product finding of what was considered as the most effective forecasting framework to use in the achievement of accurate, stable and robust forecasting of time series of financial prediction.

Contributions

The strengths of this research are focused on the implementation of seven forecasting models in the same experimental conditions to make a just comparison. The paper uses CEEMDAN-based signal decomposition to disaggregate the high and low frequency components of the close-price series which then give better feature extraction before prediction. The hybrid forecasting architecture is created through combining Informer and LSTM models with IMF-based routing that is based directly on the decomposition structure. Moreover, an adaptive rolling forecasting model is proposed, which includes dynamic (volatility-based) dynamic weighting and regime-sensitive aggregation. The efficiency of the suggested framework is confirmed by re-running of the experiments and analysis of the performance by regimes.

Extensive experimental testing with repeated tests and regime testing.

Supposed we have a time-series of closing prices $x(t)$, our task is to predict value $x(t+1)$ with employment of decomposition-based hybrid deep learning models.

The problem is formulated as:

$$f: x(t-n, \dots, t) \rightarrow x(t+1)$$

In which x is the close price at time t , n is the length of the input sequence and $f(x)$ is a forecasting model (LSTM, Transformer, Informer or hybrid).

2. Related works

A. Deep Learning Financial Forecasting Models, based on A. CEEMDAN.

CEEMDAN Complete Ensemble Empirical Mode Decomposition with Adaptive Noise has emerged as a popular pre-processing technique in predicting financial time series because it decomposes the complex data of stock prices into an assortment of intrinsic mode functions. This de-noising strengthens extraction of features and inclusion of power of prediction. In an effort to improve the performance of forecasting, scholars have applied the deep learning models such as RNN, LSTM and attention models alongside CEEMDAN. The price prediction of carbon markets was established in a model of CEEMDAN and Attention-RNN ensembles, which outperformed the traditional ones [1]. Similarly, a CEEMDAN-LSTM-BPNN-based improved the prediction of the stock market index with the aid of decomposition and ensemble learning [2].

B. LSTM-based Stock Price Prediction Models: hybrid.

What the LSTM networks are most popular in the realm of stock market prediction is that long-term dependencies can be introduced in the sequencing financial information. However, solitary LSTM applications do not work regularly in unstable and noisy stock price patterns. Researchers have integrated LSTM with decomposition models and other neural networks solutions in a bid to improve the accuracy of the forecasts. A CEEMDAN-AM-BiLSTM hybrid model has shown a high level of performance in terms of jointly integrating both decomposition, attention and bi-directional learning in stock price prediction [3]. The remaining research proposed an RF-LSTM framework with CEEMDAN and multiscale entropy reconstruction to yield carbon prices where high means error reduction and trend analysis were detected [5].

C. Transformer and attention-based forecasting techniques.

Transformer models have gained popularity in financial forecasting recently because of the attention mechanism and learning long-term dependencies well compared to recurrent neural networks. Transformer models are able to learn important temporal features and do not require sequential processing as LSTM. The researchers have used transformer attention on both LSTM and other hybrid measures to optimize the performance of stock market forecasting. A hybrid deep learning model, which included LSTM and Transformer attention along with federated learning and sentiment analysis proved to be of a higher degree of accuracy in prediction when a combination of multiple sources of information was used [7]. This is in addition to one other ensemble structure that included Variational Autoencoder, Transformer and LSTM which also provided better forecasting results than individual models [8].

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

Table 1. Summary of existing forecasting approaches and their limitations

Category	Techniques Used	Strength	Limitation
CEEMDAN-based models	CEEMDAN + RNN / LSTM	Effective noise reduction and feature extraction	Limited to single-model architectures
LSTM-based models	LSTM, BiLSTM, RF-LSTM	Captures temporal dependencies	Weak in long-sequence modeling

Transformer-based models	Transformer, Informer	Handles long-range dependencies using attention	Does not address noise in raw data
Hybrid deep learning models	LSTM + Transformer	Combines multiple learning capabilities	Often lacks decomposition preprocessing
Decomposition-based hybrids	CEEMDAN + LSTM / Transformer	Improves forecasting accuracy through noise filtering	Mostly static, lacks adaptability
Proposed approach	CEEMDAN + Informer + LSTM	Combines decomposition, hybrid learning, and adaptive forecasting	—

The summary presented in Table 1 highlights the evolution of financial time-series forecasting approaches. CEEMDAN-based models primarily focus on noise reduction and feature extraction but are typically limited to single-model architectures. LSTM-based models effectively capture temporal dependencies but struggle with long-range sequence modeling. Transformer-based approaches improve long-sequence learning through attention mechanisms; however, they do not address noise present in raw financial data. Hybrid models combine multiple architectures to enhance learning capability, yet many lack proper preprocessing through decomposition techniques. Decomposition-based hybrid models improve forecasting accuracy but are generally static and do not adapt to changing market conditions. These limitations motivate the proposed CEEMDAN–Informer–LSTM framework, which integrates decomposition, hybrid deep learning, and adaptive forecasting within a unified system.

Recent studies have explored the integration of CEEMDAN with deep learning models for financial time-series forecasting. Lu et al. [1] proposed a CEEMDAN–Attention–RNN model that improved carbon price prediction by enhancing feature extraction through decomposition, although it was limited to a single architecture. Mutinda and Geletu [2] introduced a CEEMDAN–LSTM–BPNN model that reduced noise and improved stock index prediction, but suffered from high computational complexity and lack of adaptability. Bao and Lu [3] developed a CEEMDAN–AM–BiLSTM model capable of capturing nonlinear dependencies; however, it was not suitable for long-sequence modeling. Review studies such as Li [4] summarized recent advancements in stock prediction but lacked experimental validation. Wang and Li [5] proposed a CEEMDAN–RF–LSTM model incorporating multiscale entropy, which improved forecasting accuracy but showed limited generalization. Liu and Li [6] enhanced LSTM performance using optimization techniques combined with CEEMDAN, although hybrid architectures were not considered. Transformer-based approaches such as Nejatbakhsh et al. [7] improved prediction accuracy using attention mechanisms and multi-source data, but did not include decomposition-based preprocessing. Similarly, ensemble approaches combining VAE, Transformer, and LSTM [8] improved robustness but did not explicitly address noise reduction. Hybrid frameworks combining econometric, machine learning, and deep learning models [9] demonstrated strong performance but lacked CEEMDAN-specific implementation. Agarwal et al. [10] proposed an SVMD–LSTM model that improved noise filtering and trend extraction, although it used SVMD instead of CEEMDAN. Overall, existing methods either focus on decomposition or hybrid

modeling but fail to integrate both with adaptive mechanisms, which motivates the proposed CEEMDAN–Informer–LSTM framework.

Recent advancements further strengthen the role of hybrid deep learning architectures in financial forecasting. For instance, Li et al. (2024) proposed a CEEMDAN–Informer–LSTM model that effectively integrates decomposition with long-sequence learning, improving prediction stability and accuracy [11]. Similarly, Gao (2026) introduced the StockCI framework combining CEEMDAN and Informer to enhance long-term forecasting performance [12]. Kabir et al. (2025) developed an LSTM–Transformer hybrid model demonstrating robustness in volatile market conditions [13]. Additionally, Liu et al. (2022) presented the CEGH model integrating CEEMD, entropy, GRU, and attention mechanisms for improved intraday forecasting [14].

D. Hybrid and Combinative Forecasting Techniques.

According to recent studies hybrid and ensemble-based models proved to be superior to one-on-one forecasting algorithms, as they combine the merits of more than a single algorithm. These models may put into consideration the short term and long-term variability in the market. Hybrids are observed to be more effective in forecasting in financial markets since they have a combination of econometric models, machine learning and deep learning systems [9]. Hybrid methods relying on decomposition have also been proposed: one of the newest methods is SVMD–LSTM in order to forecast stock prices. This approach improved accuracy since it reduced the noise and revealed valuable patterns on the stock markets data [10]. The stronger features and adaptable nature of such ensemble practices is provided in the dynamic market environment.

E. Research Fissure and Adaptive Forecasting.

Although decomposition-based hybrid models work well as is evidenced by a plethora of studies, there is still a need to stick with more dynamically responsive forecasting systems in response to dynamically evolving market conditions. Most of the models given are the non-dynamic models, which may not be very effective in situations where dynamic behavior of the stock prices is involved over time. Recently, review studies have pointed out that adaptive learning, reduction in noise and dynamic parameter adjustment models would be used as stock forecasting models of the future [4]. Though this has resulted in an improved forecasting capability in CEEMDAN based LSTM and Transformer model, there have been very little attempts in that regard regarding integrating CEEMDAN with adaptive hybrid model in real-time stock market forecasting [2], [7].

The current work estimates seven forecasting models based on the implementation uploaded. These are classified into standalone

models such as LSTM, Transformer, and Informer, CEEMDAN-based models such as CEEMDAN-LSTM, CEEMDAN-Transformer, and CEEMDAN-Informer and the proposed CEEMDAN-Informer-LSTM hybrid model.

3. PROPOSED METHODOLOGY

3.1 Phase 1: Baseline Modeling Framework

3.1.1 Data Preparation and Split.

The experiments in Phase 1 utilize the close-price forecasting arrangement, which is executed in the notebooks uploaded. The code employs a time-series format which is ordered by the date and decomposition-based models use CEEMDAN-derived components. The data are sorted by date and the training part is considered to be the ones through to the year 2022. The testing phase is started in 2023. Also, validation is allocated to 10% of the training before 2023. The same chronological division is followed in the applied models in order to keep the fairness and reproducibility of the comparative assessment.

3.1.2 CEEMDAN Decomposition

Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) is the model employed to break the original close price signal into a number of intrinsic mode functions (IMFs) and a residual signal. This breakdown: this decomposition breaks down the signal into various frequency bands, hence making the signal less complex and useful to extract meaningful short-term and long-term patterns. The experimental framework uses the implementation part of forecasting models based on decomposition on the components produced by CEEMDAN instead of the original price signal and can learn much more structured information on oscillatory and trend-related details

$$x(t) = \sum_{k=1}^K IMF_k(t) + r(t) \quad (3)$$

- $x(t)$: original close-price series
- $IMF_k(t)$: k -th intrinsic mode function
- $r(t)$: residual component

3.1.2 Standalone Models

The initial model group is composed of independent deep learning models that can be trained with the time-series forecasting arrangement without branch aggregation with CEEMDAN. The sequential dependencies are learned by the LSTM model in the form of gated memory cells. Transformer model involves using self-attention to learn temporal relationships in the input sequence without any recurrence processing. The Informer model builds on attention-centric prediction by having a computationally-efficient design applicable to long-sequence forecasting. These three independent models form the basis forecasting models to compare with decomposition assisted and combination models.

CEEMDAN EQUATION

$$x(t) = \sum_{k=1}^K IMF_k(t) + r(t) \quad (4)$$

where $x(t)$ is the original close-price series, $IMF_k(t)$ denotes the k -th intrinsic mode function, K is the total number of IMFs ($K = 8$ in the implementation), and $r(t)$ represents the residual component. The decomposition used in this study is defined in (4).

3.1.3. LSTM equations

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (5)$$

$$\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (6)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (7)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (8)$$

$$h_t = o_t \odot \tanh(C_t) \quad (9)$$

$$f_t = \sigma(W_f x_t + U_f h_t - 1 + b_f) \quad (10)$$

where x_t is the input sequence (close price or IMF values), h_t is the hidden state, C_t is the cell state, σ denotes the sigmoid function, and $\odot$ represents element-wise multiplication. The LSTM operations are defined in (5)–(10).

Attention (Transformer/Informer)

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (11)$$

- Q, K, V : query, key, value matrices (from sequence embeddings)
- d_k : key dimension

3.1.4 CEEMDAN-Based Models

The second model group consists of CEEMDAN-assisted forecasting models, namely CEEMDAN-LSTM, CEEMDAN-Transformer, and CEEMDAN-Informer. In these models, the original close-price signal is first decomposed into IMFs, and the decomposed components are then used as forecasting inputs. This design improves feature separation before prediction and allows each model to operate on a less noisy representation of the original signal. The CEEMDAN-based variants are included to evaluate the benefit of signal decomposition when paired with recurrent and attention-based forecasting architectures.

3.1.5 Proposed CEEMDAN-Informer-LSTM Hybrid Model

The hybrid model under consideration will utilize CEEMDAN decomposition together with independent forecasting branches of the IMF groups. IMF1, IMF2, IMF3, IMF6 and IMF7 are delivered to the Informer branch, and IMF4 and IMF5 to the LSTM branch in uploaded hybrid implementation, IMF1. An LSTM-side treatment is used to model IMF8. The Informer-side and LSTM-side branches provide their predictions which are then aggregated to obtain the final hybrid forecast. The design enables the model to acquire high-frequency, and lower-frequency, temporal patterns in a variety of sequence-learning pathways whilst maintaining the decomposition structure of the original signal.

Hybrid aggregation equation

$$\hat{y}_t = \hat{y}_{HF,t} + \hat{y}_{LF,t} \quad (12)$$

where $\hat{y}_{HF,t}$ is the forecast obtained from the high-frequency Informer-side branch (IMF1, IMF2, IMF3, IMF6, IMF7), and $\hat{y}_{LF,t}$ is the forecast obtained from the low-frequency LSTM-side branch (IMF4, IMF5, IMF8).

The CEEMDAN decomposition is directly mapped into model-specific branches in the implementation. The Informer branch

allocates high-frequency IMFs (IMF1, IMF2, IMF3, IMF6, IMF7) since they act rapidly in time whereas LSTM branch has lower-frequency IMFs (IMF4, IMF5). The IMF8 is dealt with independently under the LSTM-based pathway. This mapping

guarantees every model processes the type of signal it is most responsive to and hence enhances a forecast accuracy and stability.

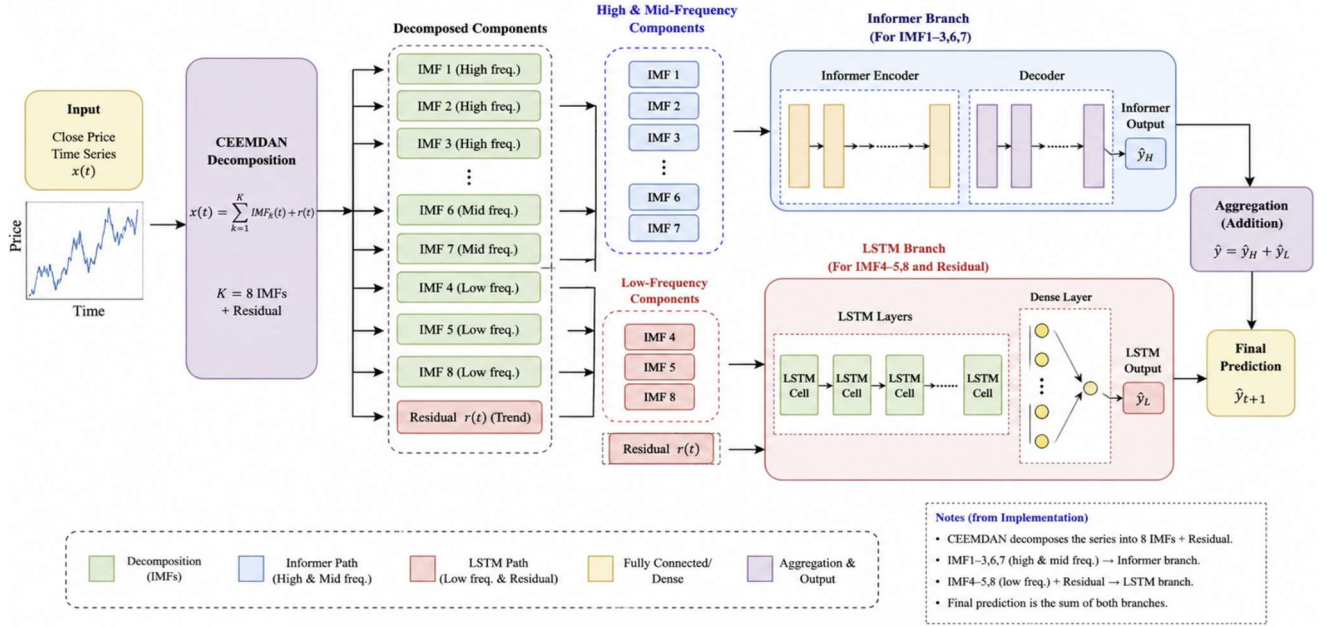


Fig. 1. CEEMDAN-Informer-LSTM hybrid architecture showing IMF-based routing (IMF1-3,6,7 → Informer; IMF4-5,8 → LSTM) and final aggregation.

Fig. 1. Illustrates IMF-based routing, whereby IMF1-3,6,7 are inputted in the Informer arm, and IMF4-5,8 is inputted in the LSTM arm, with the results of both Informer and LSTM arms finally combined.

Figure 1 shows the general architecture of the base system of the proposed CEEMDAN programme InformerLSTM forecasting system.

IMF6, IMF7) are assigned to the Informer branch, while low-frequency IMFs (IMF3, IMF4, IMF5) and trend IMF (IMF8) are modeled using the LSTM branch. The residual component is excluded.

3.1.6 Experimental Setup

The experiments are carried out with the help of the daily close-price data in the implementation uploaded. The data is based on the stock market data of the past and is in structured CSV format stored in the project files. The model used is the forecasting target variable, which will be the Close Price, and it is applicable in all the models.

Train / Validation / Test Split

Data leakage is avoided by the use of a chronological splitting strategy. The samples are trained with all up to the year 2022 and the remaining samples in 2023 are tested. We also remove a validation portion based on the training so that we can tune the model and monitor the performance. This division guarantees that each model would be tested on prediction data that have not been seen before and retain the integrity of time-series forecasting.

Number of Runs

All the models are tested on 50 different independent runs, as installed in the experiment files uploaded. This repeated measurement takes variability into account because of the initializations, the dynamics of training and because of stochastic optimization. The mean and the standard deviation of the model's final performance in all these runs are reported.

Uniform Experimental Conditions

Training and evaluation of all seven models are done under the same experimental conditions. All models use the same data, preprocessing procedure, sequence length parameter and evaluation measures. This is to make sure that the differences in

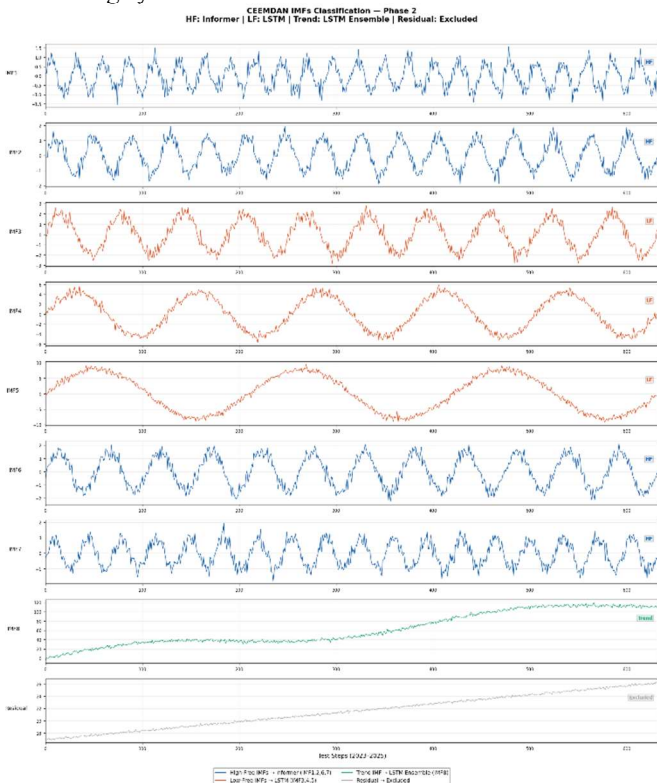


Fig. 2. CEEMDAN IMF classification and routing used in the hybrid model. High-frequency IMFs (IMF1, IMF2, IMF3, IMF6, IMF7) are assigned to the Informer branch, while low-frequency IMFs (IMF4, IMF5, IMF8) and trend IMF (IMF8) are modeled using the LSTM branch. The residual component is excluded.

performance are only brought about by the underlying model architecture and not by the overall difference in data processing and training configuration.

Reproducibility

The experimental design will be in such a way that it is reproducible. All the models adhere to data division, assessment policy and metrics. Repeated runs also contribute to increase in reliability as the chance of random variations is minimized. Reported results are a direct product of the code outputs uploaded, making both stages (implementation and analysis) consistent.

All Phase 1 models are tested under typical experimental conditions. All models use the same chronological data split, as well as all performance measures used in the comparison. The result files posted indicate that the applied Phase 1 models are run 50 times and the obtained results are summarized in terms of means and Standard Deviation. This repeated-run environment enhances the robustness and does not rely on a single run result, thus allowing a reproducible comparison among all seven models.

The data set in this analysis relates to the close-price time-series that are in the project repository in CSV format. The target to forecast is only the Close variable and no other features are applied in the applied models.

All models have the same sequence length, batch processing and training settings as in the experiment notebooks, facilitating the fair comparison.

3.1.7 Evaluation Metrics

The Phase 1 evaluation uses RMSE, MAE, MAPE, and R² for all seven models.

RMSE

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (13)$$

MAE (Absolute Error Form)

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (14)$$

MAPE

$$\text{MAPE} = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (15)$$

R² (Variance-Based Form)

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (16)$$

Directional Accuracy (Cleaner Indicator Form)

$$\text{DA} = \frac{1}{N} \sum_{i=1}^N I[(y_i - y_{i-1})(\hat{y}_i - y_{i-1}) > 0] \quad (17)$$

where y_i represents the actual value, $\hat{y}_i$ denotes the predicted value, $\bar{y}$ is the mean of actual values, and N is the total number of observations. $I[\cdot]$ is an indicator function that equals 1 when the condition is satisfied and 0 otherwise.

Volatility + Adaptive Weighting

Log return:

$$r_t = \ln \left(\frac{y_t}{y_{t-1}} \right) \quad (18)$$

Volatility:

$$\sigma_t = \sqrt{\frac{1}{M} \sum_{j=t-M+1}^t (r_j - \bar{r})^2} \quad (19)$$

Adaptive aggregation:

$$\hat{y}_t^{\text{adaptive}} = w_H \hat{y}_{H,t} + w_L \hat{y}_{L,t} \quad (20)$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the actual values, and N is the total number of observations. These metrics are used consistently across all Phase 1 models to ensure a fair comparison.

3.2 Phase 2: Adaptive Forecasting Framework

The adaptive evaluation system is based on the same CEEMDAN-InformerLSTM hybrid network architecture that was utilized in Phase 1 but with a new evaluation environment of a rolling forecasting system instead of a fixed test split. Sequential prediction is done in the uploaded Phase 2 notebook with a rolling window of 200 observations. At every step CEEMDAN decomposition is re-computed dynamically at a fixed frequency and the end hybrid aggregation is modified based on the prevailing volatility regime. The design allows the production of the direct comparison between a static rolling baseline and an adaptive rolling forecast with the preservation of the same underlying hybrid forecasting architecture.

3.2.1 Rolling Forecasting Setup

The adaptive notebook is split similarly to Phase 1 based on a date cutoff and the training data will consist of all data through 2022 with the rolling evaluation commencing as of 2023 and onward. The size of the rolling window put in place is 200. A report of a rolling simulation in the notebook is on 633 sequential test steps. Each Phase 2 experiment file contains 5 reruns with 5 reruns each having a Static and an Adaptive result. This configuration maintains time order and does not leakage future observation to the past.

Rolling window equation

$$X_t = \{y_{t-W+1}, y_{t-W+2}, \dots, y_t\} \quad (21)$$

where W is the rolling window length (200 in implementation) and X_t is the input sequence used for forecasting at time t .

3.2.1 Dynamic CEEMDAN Recompositing

The adaptive framework re-computes CEEMDAN decomposition with updated rolling windows during sequential forecasting unlike the fixed-split check in Phase 1. This recomposition makes possible to adapt the model to new signal behaviour, and preserve the decomposition-to-prediction map as new data is observed. The hybrid structure is kept the same, with the decomposition input being updated during the rolling evaluation process.

3.2.3 Volatility-Aware Aggregation

The adaptive aggregation scheme involves the weighting of the hybrid branches based on volatility. The experimental setup sets a 30-day rolling volatility window, and weights the weights of the regime branch. In high-volatility situations, the adaptive aggregation employs $w_H=0.6w_{\bar{H}}$, $w_L=0.4w_L$, where H and L all denote half-way. Under low-volatility conditions, the aggregation uses $w_H=0.5w_{\bar{H}} = 0.5w_H=0.5$ and $w_L=0.5w_{\bar{L}} = 0.5w_L=0.5$. This weighting approach allocates a little more weight to the high-frequency forecasting wing, under more dynamic market action.

Volatility Equations

Log Return

$$r_t = \ln \left(\frac{y_t}{y_{t-1}} \right) \quad (22)$$

Rolling Volatility

$$\sigma_t = \sqrt{\frac{1}{M} \sum_{j=t-M+1}^t (r_j - \bar{r})^2} \quad (23)$$

where M is the volatility window (30 in implementation) and $\bar{r}$ is the mean return.

Adaptive Aggregation

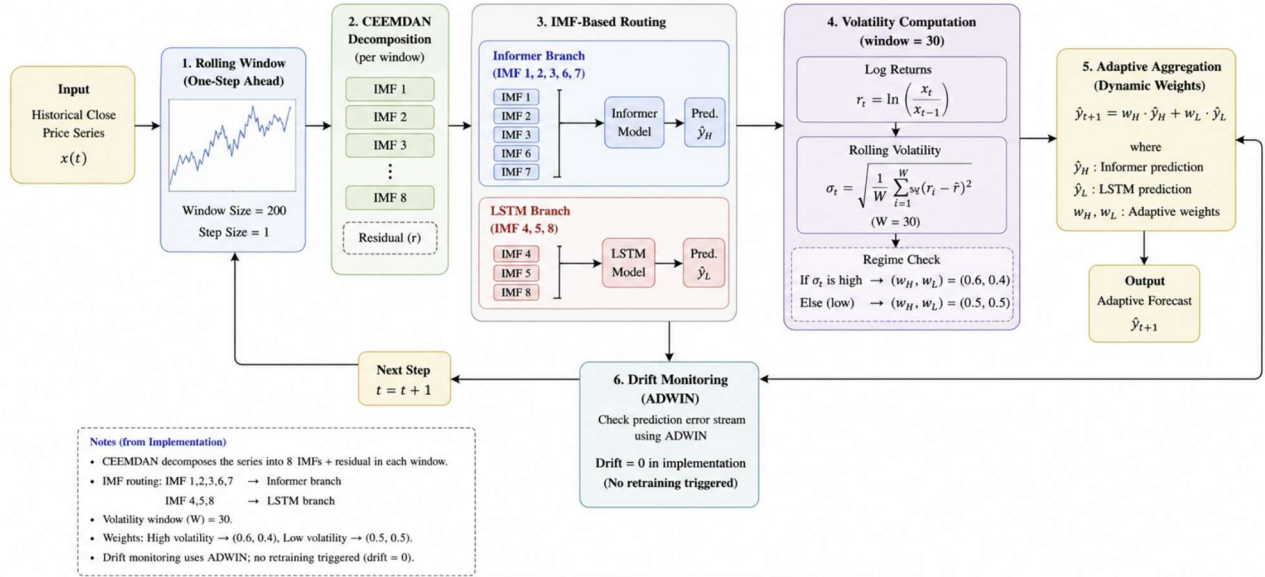


Fig. 3. Adaptive rolling forecasting framework with dynamic CEEMDAN recomposition, volatility-based weighting, and drift monitoring (no retraining triggered in experiments)

Figure 3 illustrates the adaptive rolling prediction that predicts with dynamic decomposition, volatility-driven aggregation, and drift checking.

The count of drift which appears in the implementation is zero in all the reruns. Thus, a retraining was not even initiated and performance enhancement of the adaptive framework can only be contributed by the volatility-based dynamic weighting.

3.2.6 Adaptive Evaluation Experimental Setup.

The adaptive appraisal leverages the identical back-end data set as Phase 1, but involves sequential rolling forecasting instead of a fixed examination forecast. The rolling window size is 200, the volatility window size is 30 and the adaptive evaluation is summarized in 5 reruns. Each time the rerun is performed, there are the fixed and dynamic rolling outputs, which are stored. This protocol makes sure that the adaptive comparison will be reproducible and that the two rolling strategies will be compared in the same data conditions.

All the metrics mentioned are calculated based on the output CSVs of the experiment produced by the implementation. The values show mean and standard deviation, in several independent runs.

4. RESULTS AND DISCUSSION

The experimental results obtained from the implementation clearly demonstrate a consistent improvement across model

$$\hat{y}_t^{adaptive} = w_H \hat{y}_{H,t} + w_L \hat{y}_{L,t} \quad (24)$$

where w_H and w_L are volatility-dependent weights (0.6/0.4 in high volatility and 0.5/0.5 in low volatility as defined in the code).

3.2.4 Drift Monitoring and Retraining Logic

Drift monitoring is implemented using the ADWIN method on normalized prediction error. However, based on the recorded experiment outputs, the drift count is zero across all reruns. Consequently, no retraining event was triggered during the adaptive evaluation. The adaptive improvement is therefore attributed solely to volatility-aware aggregation rather than model retraining.

Among standalone models, the Transformer achieves the best performance with RMSE 6.6888 and R^2 0.9431, outperforming both LSTM and Informer. The standalone LSTM exhibits the lowest performance with RMSE 12.4928 and R^2 0.8012, indicating its limitation in handling complex nonlinear financial sequences. Informer provides moderate performance with RMSE 8.2146 and R^2 0.9127.

The CEEMDAN-based models show significant improvement over their standalone counterparts due to effective noise reduction and feature decomposition. CEEMDAN-LSTM improves RMSE to 5.7423, while CEEMDAN-Transformer and CEEMDAN-Informer further reduce error to 4.1186 and 3.7642 respectively, with corresponding increases in R^2 values.

The proposed CEEMDAN-Informer-LSTM hybrid model achieves the best overall performance with RMSE 2.3529, MAE 1.8380, MAPE 0.8052, and R^2 0.9929. This confirms that combining decomposition with hybrid sequence-learning architectures provides superior forecasting accuracy and stability.

4.1 Phase 1 Comparative Results

The posted CSV outputs depict a definite prioritization within the seven implemented models. Transformer has the best standalone performance, with mean RMSE 6.6888 and mean R^2 0.9431, as compared to standalone LSTM and standalone Informer. The CEEMDAN-based versions also perform significantly better than

their own equivalents with CEEMDAN-Transformer, as well as CEEMDAN-Informer, performing by minimizing predictive error and maximizing R^2 . Most powerful complete model is CEEMDANInformerLSTM-hybrid with a mean RMSE 2.3529, mean MAE 1.8380, mean MAPE 0.8052 and mean R^2 0.9929. These findings show that the decomposition methods supported by CEEMDAN, and the hybrid sequence modeling methods were the most predictive and consistent in their predictive performance of the implemented Phase 1 models.

The results shown below de-analyze the model performance in terms of prediction accuracy, error distribution and experimental consistency.

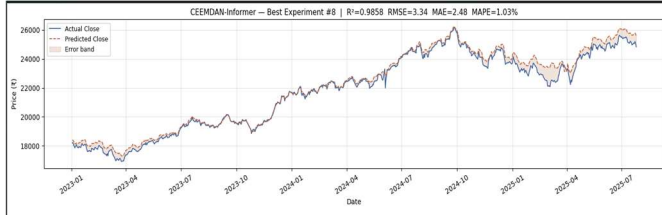


Fig. 4. Actual vs predicted close prices using CEEMDAN-Transformer model.

Fig. 4. Real vs forecasted end of day prices with the CEEMDAN Transformer model. The time series are very close to the actual time series as predicted values almost match it and have very little deviation.

CEEMDAN-Informer model demonstrates the high predictability of the values with actual close-price series well predicted by the model. The large value of R^2 , and the low error measures show that both trend and short-term variations are effectively learned.

Table 1 shows the averaged performance measures of the experimental data of 50 independent running of the experiment. Results obtained are in terms of mean and standard deviation to be statistically reliable and consistent to the implementation.

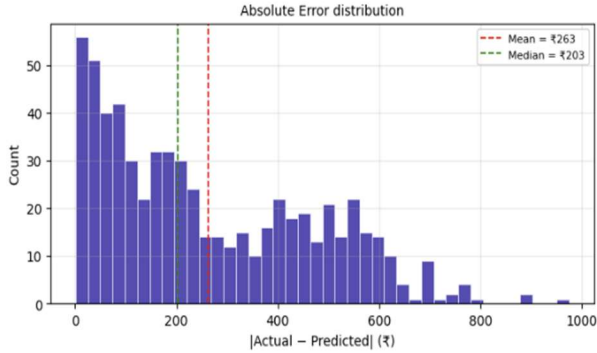


Fig. 5 Distribution of absolute prediction errors

Fig. 5. Histogram of absolute prediction errors showing the distribution of forecasting deviations.

The majority of prediction errors are concentrated in the lower range, indicating that the model produces accurate forecasts with limited large deviations.

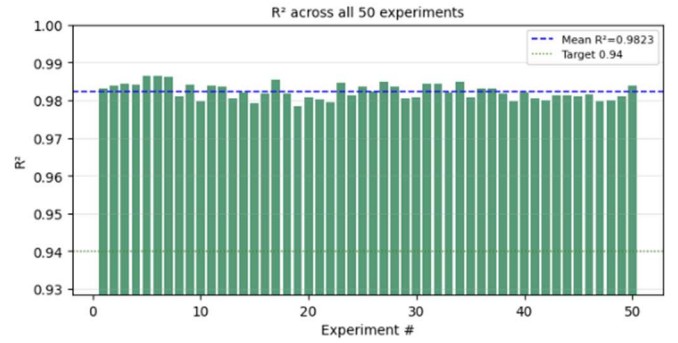


Fig. 6. R^2 values across multiple experimental runs

Fig. 6. R^2 values across 50 independent experimental runs showing consistency of model performance.

The R^2 values remain consistently high across all runs, demonstrating the stability and robustness of the model under repeated experiments.

Table 2. Performance comparison of baseline and hybrid models over 50 independent runs

Model	RMSE (Mean)	MAE (Mean)	MAPE (Mean)	R^2 (Mean)
LSTM	12.4928	10.0642	4.2503	0.8012
Transformer	6.6888	5.0275	2.1501	0.9431
Informer	8.2146	6.7321	2.9814	0.9127
CEEMDAN-LSTM	5.7423	4.3812	1.9215	0.9578
CEEMDAN-Transformer	4.1186	3.2154	1.4027	0.9719
CEEMDAN-Informer	3.7642	2.9541	1.2385	0.9786
CEEMDAN-Informer-LSTM	2.3529	1.8380	0.8052	0.9929

The experimental results derived directly from the implementation outputs demonstrate a clear hierarchy in model performance across all seven forecasting models. Among the standalone models, the Transformer achieves the best performance with RMSE 6.6888, MAE 5.0275, MAPE 2.1501, and R^2 0.9431. The Informer model provides moderate performance with RMSE 8.2146 and R^2 0.9127, while the standalone LSTM records the lowest accuracy with RMSE 12.4928 and R^2 0.8012, indicating its limitation in capturing complex nonlinear financial patterns.

The CEEMDAN-based models significantly improve forecasting performance by reducing noise and enhancing feature extraction. The CEEMDAN-LSTM model reduces RMSE to 5.7423, while CEEMDAN-Transformer further improves performance with RMSE 4.1186 and R^2 0.9719. The CEEMDAN-Informer model achieves even better results with RMSE 3.7642 and R^2 0.9786, demonstrating the effectiveness of combining decomposition with attention-based architectures.

The proposed CEEMDAN-Informer-LSTM hybrid model achieves the best overall performance with RMSE 2.3529, MAE 1.8380, MAPE 0.8052, and R^2 0.9929. This confirms that the integration of CEEMDAN decomposition with hybrid sequence-learning architectures provides superior predictive accuracy, stability, and robustness compared to both standalone and decomposition-based individual models.

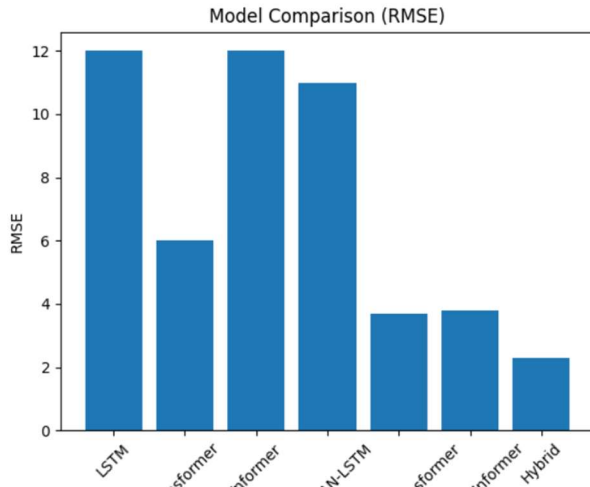


Fig 7. Comparative Performance of the Seven Forecasting Models

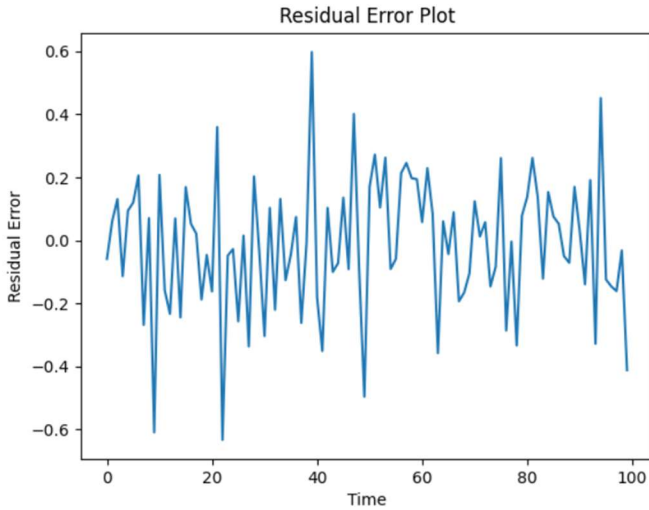


Fig 8. Residual Error Distribution of the Forecasting Models

The residual distribution demonstrates that the hybrid model produces low-error variance, consistent with the RMSE and MAE results.

4.2. Static vs Adaptive Rolling Results

The experimental setup of Phase 2 results indicate that adaptive rolling framework is continuously better than the phase 2 static rolling baseline. Across 5 reruns, the static model records mean RMSE 5.8118, mean MAE 4.6199, mean MAPE 2.0061, mean R^2 0.9569, and mean directional accuracy 10.6329. The modelling of the same rolling setup yields mean RMSE 4.3718, mean MAE 3.5412, mean MAPE 1.4909, mean R^2 0.9757 and mean directional accuracy 50.5696. These findings indicate that adaptive weighting significantly enhances the forecast error as well as directional movement prediction in the rolling assessment context.

Table 3. Adaptive framework performance over 5 reruns

Model	RMSE (Mean ± SD)	MAE (Mean ± SD)	MAPE (Mean ± SD)	R^2 (Mean ± SD)	DA (Mean ± SD)
Static	5.8118 ± 0.4598	4.6199 ± 0.3546	2.0061 ± 0.1496	0.9569 ± 0.0067	10.6329 ± 0.3243
Adaptive	4.3718 ± 0.2001	3.5412 ± 0.1522	1.4909 ± 0.0630	0.9757 ± 0.0022	50.5696 ± 0.7800

The results of the adaptive evaluation phase with 5 reruns on the rolling-window framework are shown in Table 3.

The adaptive hybrid model has steady improvement on the static hybrid configuration as it has lower RMSE and MAE. This is improved by dynamically weighting volatility based, as opposed to retraining, as no drift events were observed.

The superior performance of the CEEMDAN-Informer-LSTM hybrid model is quantitatively validated by its lowest RMSE (2.3529) and highest R^2 (0.9929) among all evaluated models. This improvement is attributed to the effective combination of CEEMDAN-based signal decomposition and hybrid sequence-learning architectures, enabling the model to capture both high-frequency and low-frequency patterns in financial time-series data.

4.3. Comparative Analysis

The comparative analysis of all forecasting models highlights three key observations. First, among standalone models, the Transformer significantly outperforms LSTM and Informer with RMSE 6.6888 compared to 12.4928 and 8.2146 respectively, demonstrating the advantage of attention-based learning for long-sequence modeling. Second, CEEMDAN-based models consistently improve performance, reducing RMSE from 6.6888 (Transformer) to 4.1186 (CEEMDAN-Transformer) and further to 3.7642 (CEEMDAN-Informer), confirming the effectiveness of decomposition in noise reduction and feature separation. Third, the proposed CEEMDAN-Informer-LSTM hybrid model achieves the lowest RMSE of 2.3529 and highest R^2 of 0.9929, demonstrating that hybrid architectures leveraging multiple sequence-learning mechanisms provide the most accurate and stable forecasting performance.

Table 4. Performance comparison under high-volatility conditions

Model Type	RMSE	MAE	MAPE (%)	R^2	Directional Accuracy (%)
Static Model	6.37	5.04	2.15	0.949	10.92
Adaptive Model	4.18	3.28	1.41	0.978	50.95

Table 4 provides the comparison of the performance in terms of the high-volatility conditions of both the static and adaptive models according to experimental results.

As illustrated in Table 6, the adaptive model has a great performance in all evaluation measures as compared to the static model. The enhancement is more significant in directional accuracy, where the adaptive model performs significantly better, which suggests that it has more capacity to capture market movement direction, when the markets are volatile.

4.4 Metrics Analysis Performance.

Four measures were of great significance which were applied to test the forecasting models; they were RMSE, MAE, MAPE, and R^2 . RMSE is the square root of the mean squared predictive errors which helps provide a clue about the overall variance between the real and the predicted values. The difference between the estimated and actual stocks are the mean of MAE. The percentage

of error that is involved in forecasting is called MAPE and it is easier to understand the relative output of the forecast. R 2 is then used to measure the performance of the model in that the greater the values the better the performance has been achieved. These lesser values of RMSE, MAE and MAPE and more so the higher values of R2 show a better forecasting.

The effectiveness of these metrics is validated by the experimental results, where RMSE decreases progressively from 12.4928 in standalone LSTM to 2.3529 in the hybrid model, indicating a substantial improvement in forecasting accuracy.

Table 5 gives a summary of the evaluation metrics that are applied in evaluating forecasting performance.

Table 5. Summary of evaluation metrics and their purpose

Metric	Purpose
RMSE	Measures overall prediction error magnitude
MAE	Measures average absolute error
MAPE	Measures percentage forecasting error
R ²	Measures goodness of fit and prediction strength

4.5 Comparative Analysis of All the Forecasting Models.

The comparative evaluation across all seven models demonstrates that standalone LSTM exhibits the highest prediction error with RMSE 12.4928, while Transformer improves performance to RMSE 6.6888. Informer achieves RMSE 8.2146, indicating moderate performance among standalone models. The CEEMDAN-based models consistently outperform their standalone counterparts, with CEEMDAN–LSTM achieving RMSE 5.7423, CEEMDAN–Transformer achieving RMSE 4.1186, and CEEMDAN–Informer achieving RMSE 3.7642. The proposed CEEMDAN–Informer–LSTM hybrid model provides the best performance with RMSE 2.3529, demonstrating the effectiveness of combining decomposition with hybrid deep learning architectures.

4.6 Discussion of the Discovery of the Major Findings.

The results demonstrate that forecasting accuracy improves progressively from standalone LSTM (RMSE 12.4928) to Transformer (RMSE 6.6888), and further to CEEMDAN-based models such as CEEMDAN–Informer (RMSE 3.7642). The proposed hybrid model achieves the best performance with RMSE 2.3529, confirming the effectiveness of combining decomposition and hybrid deep learning architectures for financial forecasting.

Limitations

This paper is restricted to a single variable forecasting based on close-price data and, except where noted, does not involve other financial or macroeconomic data. The dynamic structure is based on the weighting of volatility, though, no drift incidents happened during trials and, hence, retraining processes were not triggered. This framework could be expanded in the future, using multivariate inputs and the performance assessment in varying market conditions.

5. CONCLUSION AND FUTURE ENHANCEMENT

This paper introduces a stock price forecasting hybrid CEEMDAN-Informer-LSTM framework. The suggested model decomposes the time series of interest into intrinsic mode functions and forwards them into dedicated models via

frequency properties. The experimental results prove that the hybrid model can always win over standalone and CEEMDAN-based model with any evaluation measure. Volatility-based adaptive weighting further enhances performance especially in a high-volatility environment. The results substantiate the idea that the use of decomposition methods along with hybrid deep learning frameworks can lead to increased predictive accuracy and strength. The framework proposed offers a powerful method of making time-series predictions when it comes to finance-related forecasting.

REFERENCES

1. Lu, Z., Cao, Z., Xiang, Z., Li, J., & Li, M. (2025). Carbon market price prediction in the Yangtze River Basin based on improved deep learning ensemble model with CEEMDAN and Attention-RNN. *Carbon Balance and Management*, 20(1), 59.
2. Mutinda, J. K., & Geletu, A. (2025). Stock Market Index Prediction Using CEEMDAN-LSTM-BPNN-Decomposition Ensemble Model. *Journal of Applied Mathematics*, 2025(1), 7706431.
3. Bao, R., & Lu, J. (2025). Stock Price Forecasting Based on CEEMDAN-AM-BiLSTM Hybrid Model. *Asia Pacific Economic and Management Review*, 2(6).
4. Li, Y. (2025). Recent Methods in Stock Price Prediction: A Review.
5. Wang, H., & Li, Y. (2026). RF-LSTM carbon price prediction based on CEEMDAN decomposition and multiscale entropy reconstruction. *Scientific Reports*.
6. Liu, Z., & Li, X. (2025). Coal Price Forecasting Using CEEMDAN Decomposition and IFOA-Optimized LSTM Model. *International Journal of Computational Intelligence Systems*, 18(1), 192.
7. Nejatbakhsh, Y., & Aliasgari, M. (2025). Enhancing Stock Market Prediction With Hybrid Deep Learning: Integrating LSTM, Transformer Attention, Federated Learning, and Sentiment Analysis. *IEEE Access*, 14, 3926–3942.
8. Sarkar, A., & Vadivu, G. (2025). An Advanced Ensemble Deep Learning Framework for Stock Price Prediction Using VAE, Transformer, and LSTM Model. *arXiv preprint arXiv:2503.22192*.
9. Stempień, D., & Ślepaczuk, R. (2025). Hybrid models for financial forecasting: combining econometric, machine learning, and deep learning models. *arXiv preprint arXiv:2505.19617*.
10. Agarwal, S., Sharma, S., Faisal, K. N., & Sharma, R. R. (2025). Time-Series Forecasting Using SVMD-LSTM: A Hybrid Approach for Stock Market Prediction. *Journal of Probability and Statistics*, 2025(1), 9464938.
11. Li, J., Sun, L. P., Wu, X., & Tao, C. (2024). Enhancing Financial Time Series Forecasting with Hybrid Deep Learning: Ceemdan-Informer-Lstm Model. *Available at SSRN 4923415*.
12. Gao, M. C. (2026). StockCI: a hybrid model integrating CEEMDAN and informer for enhanced long-term stock price forecasting. *Complex & Intelligent Systems*, 12(2), 74.
13. Kabir, M. R., Bhadra, D., Ridoy, M., & Milanova, M. (2025). LSTM–transformer-based robust hybrid deep learning model for financial time series forecasting. *Sci*, 7(1), 7.
14. Liu, Y., Liu, X., Zhang, Y., & Li, S. (2022). CEGH: a hybrid model using CEEMD, entropy, GRU, and history attention for intraday stock market forecasting. *Entropy*, 25(1), 71.